

Analysis group meeting

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24 March, 2025



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- Cooking for pass0v9_dcv4 is done.
- Wait until Florian ask me to do a full pass0.

- Chris made the modifications:
 - Use a single histogram for the all runs
 - take the weighted mean
- He sent me an email on Thursday 13th, but I have miss it.
- I have cross check the results.
- One issue: the timelines have been made with the pass0v9, but it have since been deleted.
- So I have reproduce the timelines with pass0v9_dcv4 (just the values, they are not on clas12mon).

π^+ Kinematics means

run 018335	My values	Timelines
p	2.615162912	2.615163
pT	0.7197676892	0.719768
z	0.4878079392	0.487808
theta	0.3130591091	0.313059
phiH	0.1077523813	0.107752

run 018535	My values	Timelines
p	2.219703084	2.219703
pT	0.7979822965	0.797982
z	0.5041822348	0.504182
theta	0.4216313085	0.421631
phiH	0.08645434165	0.086454

π^- Kinematics means

run 018335	My values	Timelines	run 018535	My values	Timelines
p	2.477482668	2.477483	p	2.241360002	2.24136
pT	0.8316007828	0.831601	pT	0.6298523543	0.629852
z	0.4822937213	0.482294	z	0.4782910727	0.478291
theta	0.3865107356	0.386511	theta	0.3260324939	0.326033
phiH	-0.1230041012	-0.123004	phiH	-0.063859574 05	-0.06386

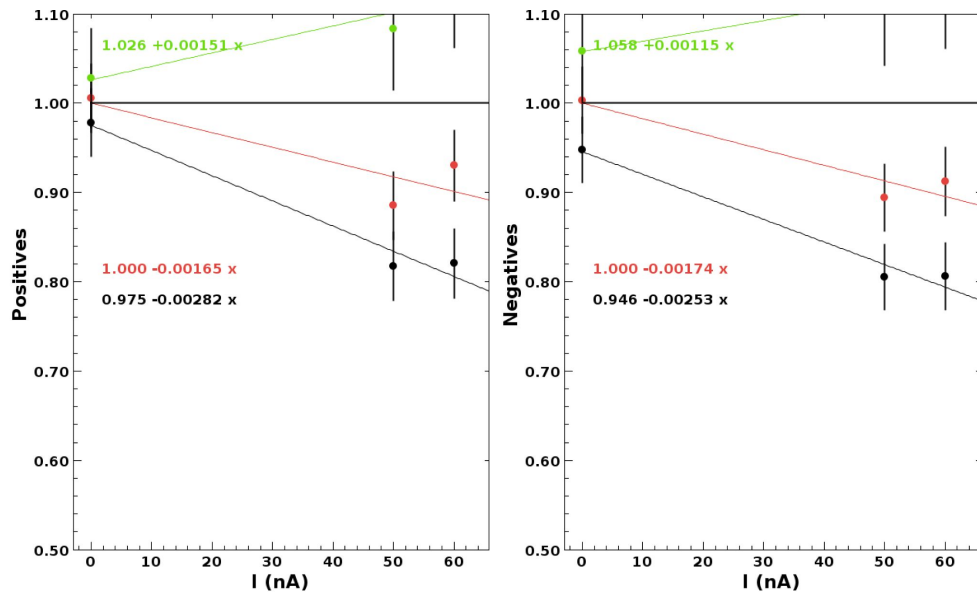
Q2 W x y Kinematics means

run 018335	My values	Timelines
Q2	2.561387695	2.561388
W	2.860547582	2.860548
x	0.2680846817	0.268085
y	0.5109740378	0.510974

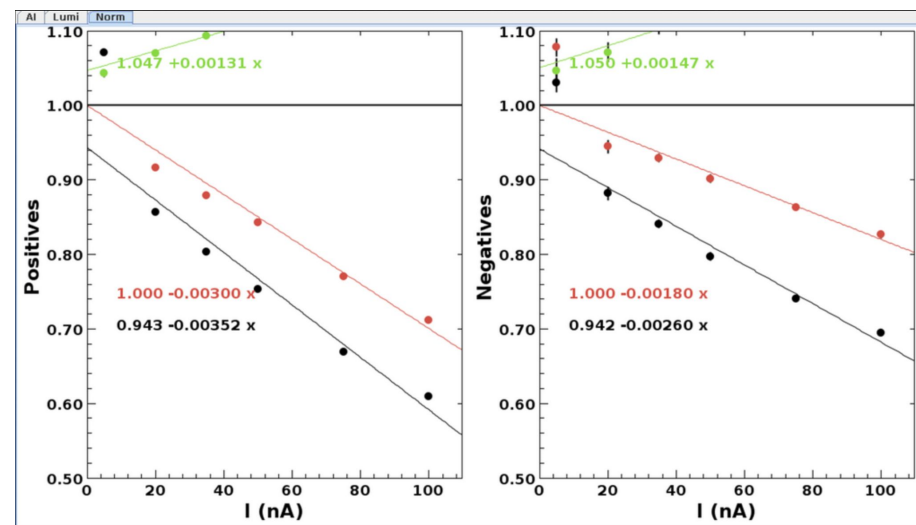
run 018535	My values	Timelines
Q2	1.785937872	1.785938
W	2.751860939	2.751861
x	0.2178727032	0.217873
y	0.441548579	0.441549

Exact match between my values and the timelines values

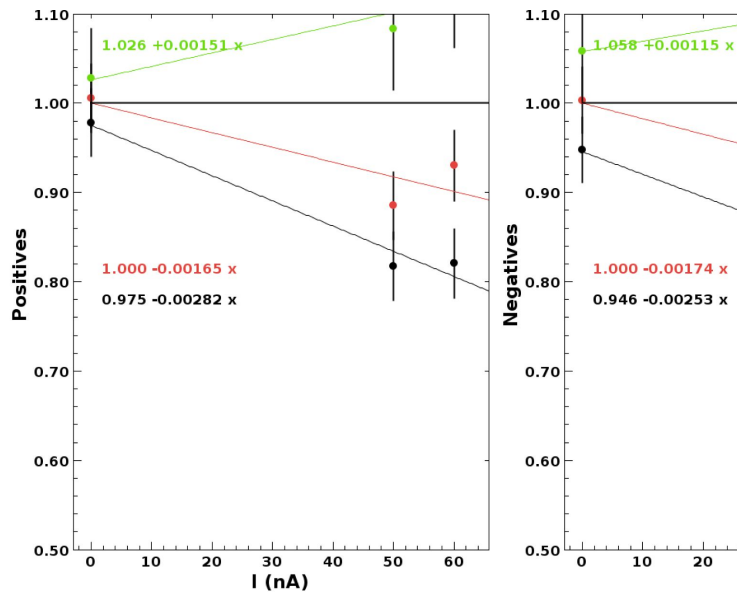
MC + background



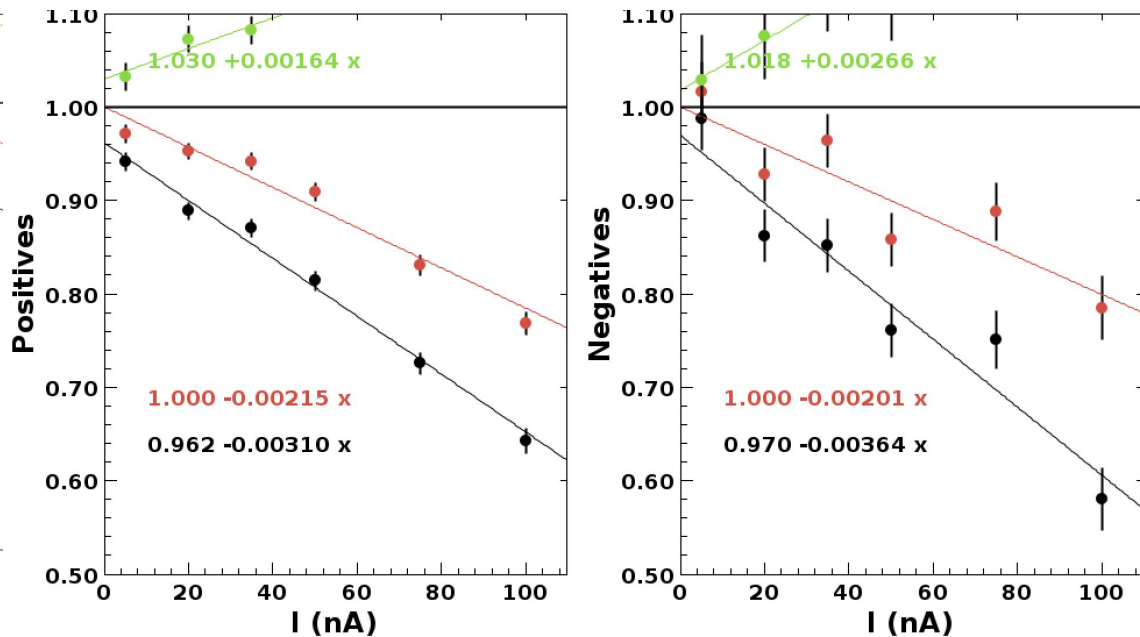
Luminosity scan (done by Daniel)



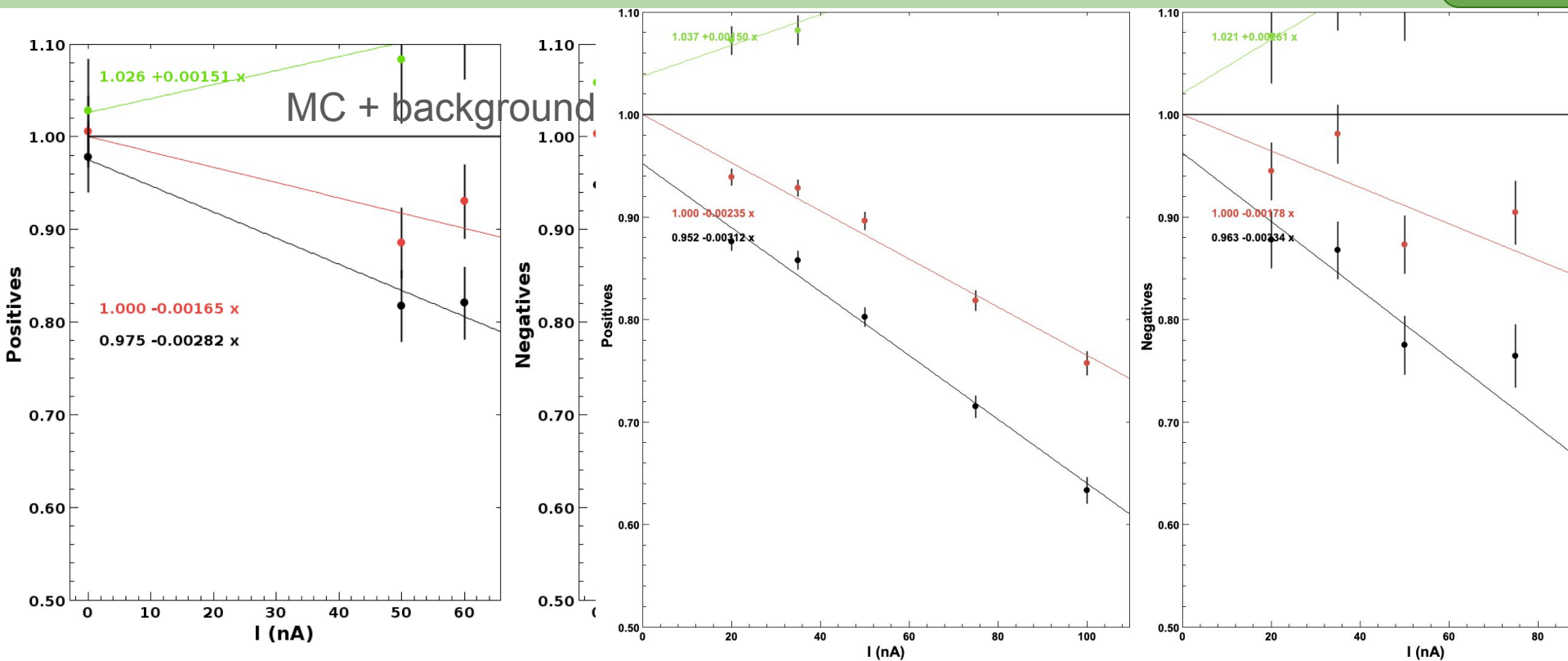
MC + background



Luminosity scan (done by me)



Background merging



I have discussed this with Raffaella, and she mentioned that the discrepancy is larger than she expected.

This might be due to the luminosity scan being processed with a different version of COATJAVA and calibration.

She advised me to redo the luminosity scan plot using the same COATJAVA version employed for the simulation and same calibration.

I launched the reconstruction jobs on the farm on Friday, but they failed due to time limits.

I relaunched them this morning, processing only 100k events per run.

Whitney found a issue (before GHP) in the function that generate all the track candidates. If the AHDC have too much hits, this function take a very long time and crash.

I started implementing a fix in COATJAVA.

Count on the number of track candidates generated.

If this counter is greater than a specific number (10 000) then the event is rejected.

```
superpreclusters_s1: [[5]]  
superpreclusters_s2: [7]  
superpreclusters_s3: [17]  
superpreclusters_s4: [18]  
superpreclusters_s5: [20, 21, 22]
```

```
combinations_s1_s2  
c: [5, 7]  
combinations_s1_s2_s3  
c: [5, 7, 17]  
combinations_s1_s2_s3_s4  
c: [5, 7, 17, 18]
```

```
combinations_s1_s2_s3_s4_s5  
c: [5, 7, 17, 18, 20]  
c: [5, 7, 17, 18, 21]  
c: [5, 7, 17, 18, 22]
```

- RG-D analysis:
 - Cross check with Matt the nuclear transparency
 - Understand the lc vs TA dependence
 - Finish the event mixing method (difference at low invariant mass)
 - Extract the multiplicity ratio, pT broadening and asymmetry
- AI for ALERT:
 - Finish the implementation of the fix
- For the shift, during the last collaboration meeting, Patrick say it is a good idea to have the Basic Electrical (ESC001) training. The next session is [3/31/2025 9:00 am](#) to 11:30. Is it fine if I miss the HallB meeting and the ALERT software meeting?